AdifySkills

n8n Automation Mastery
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📘 CLASS NOTES

🧩 Module 2 · Core Nodes & Data Mastery

DAY 05 of 25

# Logic & Flow Control: IF, Switch, Merge, Loop

"Workflows jo khud decision lete hain."

⚡ AAJ HUM YEH BANAYENGE

## Lead Scoring + Smart Routing

Webhook par lead aata hai, Code node score banata hai (budget, urgency, source), aur Switch hot lead ko Slack, warm ko nurture email, cold ko archive sheet mein bhejta hai.

IF | Switch | Merge | Loop Over Items | Wait

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

# 🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

**💡 Yeh lesson kyun zaroori hai?**

Real business processes seedhe nahi hote — "agar yeh, to woh". Flow control se n8n decisions leta hai.

## 🎯 Aaj ke baad aap yeh kar paoge

**01 IF node & conditions (AND/OR)**

🌍 Paise hain AUR dukaan khuli hai → shopping. (AND); chai YA coffee → kuch bhi chalega (OR).
⚙️ IF: {{ $json.paid }} is true AND amount > 0 → invoice branch.

**02 Switch node — multi-route routing**

🌍 Post office sorting: har letter pin code ke hisaab se alag bag mein.
⚙️ Switch on city: Delhi → rep A, Mumbai → rep B, fallback → manager.

**03 Merge node modes: append, combine, by key**

🌍 Do registers ko roll number se milana.
⚙️ Merge (matching fields: email) → CRM data + Sheet data ek item mein.

**04 Loop Over Items (batching) & rate limits**

🌍 Lift mein ek baar 8 log — baaki agli baar.
⚙️ Loop Over Items batch 50 + Wait 1s → API rate limit safe.

**05 Wait node — delays & resume on webhook**

🌍 Doctor: "3 din baad wapas aana".
⚙️ Wait 3 days → follow-up email; ya Wait "on webhook" → approval ke baad aage.

**06 Filter, Remove Duplicates, Sort & Limit**

🌍 Chhalni se kankad nikaalna, duplicate photos delete, marks ke hisaab se rank, top 10.
⚙️ Filter (city = Pune) → Remove Duplicates (email) → Sort (score desc) → Limit 10.

## ⏱ Class ka plan (2 ghante)

| 10m | 35m | 45m | 20m | 10m |
|---|---|---|---|---|

**0:00-0:10 Recap**

🪄 *Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

**0:10-0:45 Concept: Flow Control**

Flowchart → n8n mapping

🪄 *Traffic police: green to chalo, red to ruko (IF). Chauraha: left, right, seedha (Switch).*

**0:45-1:30 Live Build**

Smart lead router: hot/warm/cold → different actions

🪄 *Aaj ki kahani yaad hai — 'Railway junction — kaun si train kis platform'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

**1:30-1:50 Challenge**

Batch 500 records with rate limit

🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

**1:50-2:00 Q&A + Homework**

🪄 *Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

# 🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

### IF

2 raaste: true / false.

🪄 **Analogy:** Traffic signal: green ya red

CONCEPT 02

### Switch

3+ raaste, value ke hisaab se.

🪄 **Analogy:** Railway junction

CONCEPT 03

### Merge

Alag branches ka data wapas jodna.

🪄 **Analogy:** Do nadiyan milkar ek

CONCEPT 04

### Loop Over Items

Batch-by-batch processing — API rate limit ke liye.

🪄 **Analogy:** Ek baar mein 10 log lift mein

### 🔑 Aaj ka main idea

Flow control = workflow ka dimaag. IF 2 raaste banata hai (haan/naa), Switch kai raaste, Merge raaston ko wapas jodta hai, Loop bade data ko chhote batches mein chalata hai, Wait ruk ke baad mein chalata hai.

- IF: baarish ho rahi hai? → chhata lo, warna nahi
- Switch: marks 90+ A, 75+ B, baaki C
- Wait: chai ubalne tak 3 minute ruko

### 📖 Ek kahani se samjho: Railway junction — kaun si train kis platform

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| 🚆 Train aayi (lead) = Webhook | 🧮 Ticket check (score) = Code / Set | 🔀 Junction: 3 raaste = Switch | 🔥 Express → Platform 1 = HOT → Slack sales | 🐢 Passenger → Platform 3 = COLD → archive |

🪄 **Ek line mein: “Traffic police: green to chalo, red to ruko (IF). Chauraha: left, right, seedha (Switch).”**

# 🗺️ Aaj ka workflow: Lead Scoring + Smart Routing

Data left se right chalta hai. Har dabba ek n8n node hai.



**💰 Isse kitna kama sakte ho?**

Sales teams, B2B companies (lead scoring + routing) — aam taur par **₹10,000 – ₹25,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

## 🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Webhook**
   Lead aata hai: budget, timeline, source
2. 👨‍💻 **Code**
   Score calculate: budget + urgency + source points
3. 🔀 **Switch**
   score ≥ 70 HOT, 40–69 WARM, baaki COLD
4. 💼 **Slack**
   HOT → #sales channel turant
5. 📧 **Gmail**
   WARM → nurture email sequence
6. 📊 **Google Sheets**
   COLD → archive sheet

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

### 🔑 Setup (import ke baad)

- ☐ Connect: **Slack (OAuth / bot token)**
- ☐ Connect: **Gmail (Google OAuth)**
- ☐ Connect: **Google Sheets (Google OAuth)**
- ☐ Slack mein #sales channel banao (ya Slack node ko Telegram se replace karo)

### ✅ Test kaise karein

1. Webhook node → "Listen for test event" dabao
2. Neeche wala curl command chalao (ya Postman / Hoppscotch se bhejo)
3. Budget badal-badal ke teeno routes test karo

## ⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

### 😩 Pehle (manual)

| | |
|---|---|
| Har lead padh ke decide karna | 5 min |
| Sahi insaan ko forward | 3 min |
| Follow-up yaad rakhna | 5 min |
| | **13 min** |

### ⚡ Ab (n8n)

| | |
|---|---|
| Score + routing auto | |
| Follow-up Wait node se | |
| | **0 min** |

⏱ **Har lead par 13 min bache, hot lead 1 minute mein sales tak.**

## ⌨️ Aaj ka key snippet

**Copy karke n8n mein paste karo** 📋 Select karke copy karo

```
const s = ($json.budget > 30000 ? 40 : 10) + ($json.timeline === "this week" ? 30 : 5);
return { ...$json, score: s };
```

## 📦 Sample data — input se output tak

**📥 Input** 📋 Select karke copy karo

```
{
  "name": "Neha",
  "budget": 50000,
  "timeline": "this week",
  "source": "referral"
}
```

**📤 Output** 📋 Select karke copy karo

```
{
  "name": "Neha",
  "score": 85,
  "label": "HOT",
  "routedTo": "slack#sales"
}
```

# 💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

**💬 Build it with ChatGPT / Claude** 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Webhook par lead aata hai, Code node score banata hai (budget, urgency, source), aur Switch hot lead ko Slack, warm ko nurture email, cold ko archive sheet mein bhejta hai.

Flow: Webhook (POST /score-lead) → Code (Score 0-100) → Switch (Hot / Warm / Cold) → Slack · Gmail · Sheets (Route karo)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

**💬 Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "Lead Scoring + Smart Routing" (nodes: Lead Webhook, Score Lead, Route by Score, Hot → Slack Sales, Warm → Nurture Email, Cold → Archive).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

## 📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo ( `Ctrl` + `A` , `Ctrl` + `C` ).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

**Workflow JSON**
6 nodes · ready to import
`adifyskills.com/workflows/day-05-lead-scoring-router.json`

**Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/5`

## Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

| Agar yeh ho raha hai… | To yeh karo |
|---|---|
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Slack (OAuth / bot token)" connect karo. |
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo. |
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo. |
| **Webhook/Form test par kuch nahi ho raha?** | "Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai. |
| **"Sheet not found" / column khaali?** | Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye. |

### ⚠️ Yeh galtiyan sabse common hain

- ! Merge node ka galat mode — data duplicate ho jaata hai
- ! IF mein string "10" vs number 10 compare karna
- ! Loop ke andar "done" output ko connect na karna

### ⭐ Pro tips — time bachane wali tricks

- ★ Switch mein "Fallback output" on rakho — unexpected values ke liye
- ★ Wait node se "3 din baad" follow-up — server restart par bhi resume hota hai
- ★ Remove Duplicates (past executions) se same lead dobara process na ho

**Phir bhi atke? Koi baat nahi 🙂**
Screenshot + error message ke saath WhatsApp par poochho →

# 🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

### 🛠 Live lab

Lead scoring + routing workflow: score ke hisaab se Sales Slack, nurture email ya archive.

### 📝 Homework

Do data sources ko Merge (by key) karke unified customer list banao.

- [ ] Homework ho gaya

**BEGINNER**

Budget 0 wale leads filter karo

- [ ]

**INTERMEDIATE**

Round-robin: leads 3 sales reps mein baanto

- [ ]

**ADVANCED**

Wait node + approval link se manager approval flow

- [ ]

### 🧩 Aaj ka challenge — pehle khud try karo

**🎯 Agar lead ka source "referral" hai to score ke bawajood hamesha hot maano.**

**💡 Hint 1**

Yeh logic Code node mein ek extra line se ho sakta hai.

**💡 Hint 2**

return se pehle: agar source referral hai to score ko 70 ya usse upar kar do.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

# 🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

**Q1. 5 alag categories ke liye best node?**

(A) Switch
(B) Merge
(C) Wait
(D) IF

**Q2. API 1 sec mein sirf 10 requests leti hai — kya use karein?**

(A) Loop Over Items + Wait
(B) IF
(C) Merge
(D) Sort

**Q3. Do sources ko customer ID se jodna?**

(A) Filter
(B) Merge (by key/matching fields)
(C) Aggregate
(D) Split Out

**Q4. IF node ke kitne outputs?**

(A) 3
(B) Unlimited
(C) 1
(D) 2

## 📖 Aaj ke naye shabd

**Node**
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

**Webhook**
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

**API**
Do apps ke baat karne ka tareeka. Ek app sawaal bhejta hai, doosra jawab (data) deta hai.

**IF Node**
Haan/naa wala decision — condition sahi to ek raasta, galat to doosra.

**Switch**
Kai raaston mein se ek chunna — jaise traffic police kisi gaadi ko alag lane mein bhejta hai.

**Merge**
Do raaston ka data wapas ek jagah jodna.

**Rate Limit**
API ki speed limit — ek minute mein kitni requests allowed hain.

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## ✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 6

**Code Node: JavaScript for Automators**

"Jahan node khatam, wahan Code node shuru."

☐ **Day 5 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

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# 🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

**Q1 → A**
Switch

**Q2 → A**
Loop Over Items + Wait

**Q3 → B**
Merge (by key/matching fields)

**Q4 → D**
2

### ✅ Challenge ka solution

Score Lead code mein: if (b.source === 'referral') score = Math.max(score, 70);

## 📚 Official docs

- Flow logic ↗
- Merging data ↗
- Looping ↗
- Splitting with conditionals ↗

## ▶ Videos

- n8n if switch merge tutorial ↗
- n8n loop over items batching ↗

**Doubt poochho**
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